

Supplementary Tables S1–S17

Supplementary Tables S1–S17

The tables below report the reader-facing supplementary analyses for the current submission scope. Exact machine-readable files and implementation identifiers are mapped in the repository reader-facing provenance index.

Supplementary Table S1 | Analysis cohorts and denominator conventions

Analysis universe	Domain / stratum	Systems, n	Reconstruction rows	Primary use
Development benchmark	bulk	186	included in 6,343 total	codec benchmark, controls, reclassification, matching
Development benchmark	slab	68	included in 6,343 total	codec benchmark, controls, reclassification, matching
Development benchmark total	—	254	6,343	primary development analysis
External stability/descriptive set	bulk	37	included in 1,755 descriptive total	QSQ external stability/descriptive analysis
External stability/descriptive set	vacuum-containing 2D	28	included in 1,755 descriptive total	QSQ external stability/descriptive analysis
External descriptive total	—	65	1,755	external descriptive aggregate
Combined stability universe	development + external descriptive	319	not a codec-row denominator	QSQ stability-floor / eligibility summaries
Primary external confirmatory cohort	pre-specified confirmatory subset	63	1,689 retained scientific rows	untouched rate–fidelity confirmation

Caption. The study uses several deliberately distinct analysis universes. The 254-system development benchmark underlies the 6,343-row compression table and the central binary-to-three-state reclassification. QoI Stability Qualification (QSQ) summaries use 319 systems, comprising the 254 development systems plus all 65 records in the external descriptive/stability set. The primary external rate–fidelity confirmation is a pre-specified 63-system cohort with 1,689 retained scientific rows. The 65-system external descriptive aggregate contains 1,755 rows. Denominators are therefore analysis-specific and are not interchangeable.

Evidence provenance: cohort metadata, QSQ eligibility summary, external manifest and confirmatory summary; exact repository paths are mapped in the reader-facing provenance index.

Supplementary Table S2 | QSQ definition and order-preserving control

Property	Order-preserving round-trip control	QSQ perturbation
Role	Method-validation control	Active qualification procedure
Stability perturbation	float64 → float32 → float64 round trip	additive uniform noise $U(-\epsilon, +\epsilon)$
Perturbation amplitude	implicit float32 round-trip amplitude	ϵ equals that material's measured float32 L^∞ round-trip error
Seeds	deterministic; not applicable	five pre-specified seeds: 20260905, 1, 2, 3, 4
Stability floor	one deterministic response	maximum Bader response across five seeds
Bader basins	BaderKit, on-grid, re-derived	unchanged
Scientific thresholds	10^{-4} , 10^{-3} , 10^{-2} e	unchanged
Non-evaluable rule	floor $\geq \tau$	unchanged
Non-evaluable semantics	neither pass nor failure	unchanged
Reason for replacement	round trip is order-preserving and unusually benign to on-grid watershed ordering	non-order-preserving perturbation can excite the relevant partition instability
Benchmark role	method-validation control only	defines eligibility and certification statistics

Calibration evidence supporting QSQ. In the 18-material calibration panel, the order-preserving round-trip control created a median of 82 exact neighbouring ties and reassigned zero voxels in 9/18 systems. The QSQ perturbation created no exact ties and reassigned zero voxels in only 2/18 systems. Across five seeds, the material-wise log10 floor span had a median of 0.47 decades and reached 2.4 decades. Over the $\times 0.1$ to $\times 10$ amplitude sweep, the floor changed by a median of 0.76 decades (P10 0.00; P90 2.02).

Evidence provenance: QSQ method record, order-preserving control record and probe-calibration panel.

Supplementary Table S3 | QSQ eligibility by threshold and stratum

Bader tolerance, τ	Stratum	n	Eligible	Non-evaluable	Non-evaluable fraction	QSQ floor median (e)	QSQ floor P90 (e)	QSQ floor max (e)
10 ⁻⁴	overall	319	64	255	79.94%	6.525e-4	9.378e-3	2.913
10 ⁻⁴	development bulk	186	42	144	77.42%	6.729e-4	8.302e-3	2.913
10 ⁻⁴	development slab	68	4	64	94.12%	8.014e-4	9.929e-3	2.275e-2
10 ⁻⁴	external bulk	37	10	27	72.97%	4.953e-4	3.708e-2	2.146
10 ⁻⁴	external vacuum-containing 2D	28	8	20	71.43%	1.845e-4	1.462e-3	3.979e-3
10 ⁻³	overall	319	187	132	41.38%	6.525e-4	9.378e-3	2.913
10 ⁻³	development bulk	186	103	83	44.62%	6.729e-4	8.302e-3	2.913
10 ⁻³	development slab	68	40	28	41.18%	8.014e-4	9.929e-3	2.275e-2
10 ⁻³	external bulk	37	22	15	40.54%	4.953e-4	3.708e-2	2.146
10 ⁻³	external vacuum-containing 2D	28	22	6	21.43%	1.845e-4	1.462e-3	3.979e-3
10 ⁻²	overall	319	288	31	9.72%	6.525e-4	9.378e-3	2.913
10 ⁻²	development bulk	186	168	18	9.68%	6.729e-4	8.302e-3	2.913
10 ⁻²	development slab	68	61	7	10.29%	8.014e-4	9.929e-3	2.275e-2
10 ⁻²	external bulk	37	31	6	16.22%	4.953e-4	3.708e-2	2.146
10 ⁻²	external vacuum-containing 2D	28	28	0	0.00%	1.845e-4	1.462e-3	3.979e-3

Caption. QSQ determines whether the uncompressed Bader analysis is numerically identifiable at the requested tolerance before codec scoring. The non-evaluable fraction rises sharply at stricter tolerances. The development slab versus bulk contrast is not stable across thresholds and does not reproduce as a universal external structural-class effect; numerical eligibility is therefore measured per material rather than inferred from coarse system type.

Evidence provenance: QSQ eligibility summary.

Supplementary Table S4 | Codec-resolved binary-to-three-state reclassification in the 254-system development cohort

τ	Codec	Eligible	Non-evaluable	Naive pass	Naive fail	Qualified pass	Genuine eligible fail	Non-evaluable naive pass	Naive fail → non-evaluable	Fraction of naive failures reclassified
10 ⁻⁴	SPERR	46	208	61	193	39	7	22	186	96.37%
10 ⁻⁴	SZ3	46	208	49	205	38	8	11	197	96.10%
10 ⁻⁴	ZFP	46	208	119	135	46	0	73	135	100.00%
10 ⁻³	SPERR	143	111	144	110	137	6	7	104	94.55%
10 ⁻³	SZ3	143	111	140	114	136	7	4	107	93.86%
10 ⁻³	ZFP	143	111	168	86	142	1	26	85	98.84%
10 ⁻²	SPERR	229	25	212	42	208	21	4	21	50.00%
10 ⁻²	SZ3	229	25	209	45	207	22	2	23	51.11%
10 ⁻²	ZFP	229	25	233	21	225	4	8	17	80.95%

Caption. Each row summarizes 254 material–codec decisions at one Bader tolerance. The same material-level eligibility denominator applies to all three codecs because eligibility is measured on the uncompressed reference analysis. At 10⁻⁴ and 10⁻³ e, more than 93% of naive failures for every codec are reclassified as non-evaluable rather than genuine eligible failures. The presence of non-evaluable naive passes, particularly 73 ZFP cases at 10⁻⁴ e, shows that QSQ does not merely remove unfavorable codec outcomes; it invalidates both success and failure labels when the reference QoI is not independently resolvable.

Evidence provenance: codec-resolved reclassification summary.

Supplementary Table S5 | Sensitivity to omission of eligibility and inflation of the Bader threshold

S5a. Naive no-exclusion diagnostic

The table applies the requested Bader tolerance directly to all 254 development materials without QSQ eligibility qualification. n certified therefore means a naive binary pass and must not be interpreted as a stability-qualified scientific certification.

Bader threshold	Codec	n	Naive certified, n	Naive certified (%)	Median best ratio (x)	P10 (x)	P90 (x)
10 ⁻⁴ e	SPERR	254	61	24.0	4.19	3.81	13.13
10 ⁻⁴ e	SZ3	254	49	19.3	6.95	5.10	9.86
10 ⁻⁴ e	ZFP	254	119	46.9	7.55	5.74	10.86
10 ⁻³ e	SPERR	254	144	56.7	5.88	4.55	29.05
10 ⁻³ e	SZ3	254	140	55.1	13.23	8.21	20.64
10 ⁻³ e	ZFP	254	168	66.1	13.59	9.69	21.24
10 ⁻² e	SPERR	254	212	83.5	10.32	7.13	103.25
10 ⁻² e	SZ3	254	209	82.3	57.65	25.88	109.56
10 ⁻² e	ZFP	254	233	91.7	31.57	19.73	52.02

The corresponding pooled benchmark-validity audit is reported in main-text Figure 3 and Supplementary Table S4. In particular, apparent pass/fail labels in this no-exclusion analysis can be invalid whenever the reference Bader QoI is itself non-evaluable at the requested threshold.

S5b. Inflated-threshold stress test

Here the analysis threshold is multiplied by $k = 2, 5, 10$ as a post hoc robustness diagnostic. Values are pooled over the 254 development materials. The original chemical tolerance remains the pre-specified contract of record.

Original threshold	k	Codec	Certified, n/254	Certified (%)	Median best ratio (x)
10 ⁻⁴ e	2	SPERR	214/254	84.3	6.08
10 ⁻⁴ e	2	SZ3	209/254	82.3	13.04
10 ⁻⁴ e	2	ZFP	244/254	96.1	13.86
10 ⁻⁴ e	5	SPERR	241/254	94.9	7.47
10 ⁻⁴ e	5	SZ3	236/254	92.9	23.78
10 ⁻⁴ e	5	ZFP	253/254	99.6	20.02
10 ⁻⁴ e	10	SPERR	248/254	97.6	8.71
10 ⁻⁴ e	10	SZ3	246/254	96.9	35.19
10 ⁻⁴ e	10	ZFP	254/254	100.0	24.70
10 ⁻³ e	2	SPERR	219/254	86.2	6.70
10 ⁻³ e	2	SZ3	217/254	85.4	16.78
10 ⁻³ e	2	ZFP	244/254	96.1	16.13
10 ⁻³ e	5	SPERR	242/254	95.3	7.77
10 ⁻³ e	5	SZ3	236/254	92.9	24.63
10 ⁻³ e	5	ZFP	253/254	99.6	21.08
10 ⁻³ e	10	SPERR	248/254	97.6	8.83
10 ⁻³ e	10	SZ3	246/254	96.9	35.67
10 ⁻³ e	10	ZFP	254/254	100.0	25.01
10 ⁻² e	2	SPERR	236/254	92.9	10.43
10 ⁻² e	2	SZ3	234/254	92.1	60.81
10 ⁻² e	2	ZFP	253/254	99.6	32.30
10 ⁻² e	5	SPERR	244/254	96.1	10.58
10 ⁻² e	5	SZ3	241/254	94.9	65.24
10 ⁻² e	5	ZFP	253/254	99.6	34.60
10 ⁻² e	10	SPERR	248/254	97.6	11.10
10 ⁻² e	10	SZ3	246/254	96.9	70.92
10 ⁻² e	10	ZFP	254/254	100.0	37.83

Evidence provenance: threshold-sensitivity analysis, including bulk/slab-resolved rows and distributional quantiles.

Supplementary Table S6 | Re-derived Bader error relative to each material's independent QSQ stability floor

Only certified reconstruction points are included. Ratios near unity indicate that the observed reconstruction error is on the same numerical scale as the independent stability floor. This table does not estimate a material-level compression-error plateau and cannot establish plateau = floor.

Bader threshold	Codec	n	Median DeltaQ/floor	P10	P90
10 ⁻⁴ e	SPERR	39	1.232	0.543	3.203
10 ⁻⁴ e	SZ3	38	1.329	0.753	2.860
10 ⁻⁴ e	ZFP	46	1.091	0.338	3.247
10 ⁻³ e	SPERR	137	3.546	0.915	19.915
10 ⁻³ e	SZ3	136	2.783	0.902	16.587
10 ⁻³ e	ZFP	142	3.386	0.822	17.825
10 ⁻² e	SPERR	208	14.221	1.531	127.075
10 ⁻² e	SZ3	207	14.613	1.733	128.381
10 ⁻² e	ZFP	225	10.698	1.308	129.864

At the strictest certified contract, median DeltaQ/floor is only 1.09-1.33 across the three codecs, consistent with an emerging analysis-limited regime. The much broader ratios at 10⁻³ and 10⁻² e show that this floor-scale interpretation should not be generalized across the full tolerance range.

Evidence provenance: certified-error-to-QSQ-floor summary.

Supplementary Table S7 | QSQ seed and amplitude sensitivity

S7a. Eighteen-material probe calibration

Diagnostic	Calibration result	Qualification implication
Exact neighbouring ties created	median 82 under order-preserving control; 0 under same-amplitude QSQ perturbation	An order-preserving round trip is unusually benign for an order-dependent watershed
Systems with zero voxel reassignment	9/18 order-preserving control; 2/18 same-amplitude QSQ perturbation	QSQ perturbation more effectively probes the relevant partition-instability channel
Five-seed log10 floor span	median 0.47 decades; maximum 2.4 decades	A single seed is insufficient for the five-seed qualification rule
Seed-dependent eligibility at 10 ⁻⁴ e	3/18 materials	Five pre-specified seeds retained
Seed-dependent eligibility at 10 ⁻³ e	2/18 materials	Five pre-specified seeds retained
Seed-dependent eligibility at 10 ⁻² e	1/18 materials	Five pre-specified seeds retained
Floor change from x0.1 to x10 amplitude	median 0.76 decades; P10 0.00; P90 2.02	The floor is explicitly qualification-defined, not amplitude-free

S7b. Full 319-system application of the five-seed QSQ rule

Diagnostic	Full-corpus result
Systems	319
Five-seed log10 floor span, median	0.37 decades
Five-seed log10 floor span, maximum	3.82 decades
Primary-seed verdict differs from five-seed maximum at 10 ⁻⁴ e	35/319
Primary-seed verdict differs from five-seed maximum at 10 ⁻³ e	17/319
Primary-seed verdict differs from five-seed maximum at 10 ⁻² e	2/319

The 0.47-decade median / 2.4-decade maximum in S7a describe the calibration panel, whereas the 0.37-decade median / 3.82-decade maximum in S7b describe application of the five-seed rule to the full 319-system corpus. They answer different questions and are not interchangeable.

Evidence provenance: probe-calibration panel, per-seed QSQ responses and amplitude-sensitivity analysis.

Interpretation.

Tables S5-S7 are sensitivity and audit material. They support transparency of the measurement contract, but they do not redefine the primary chemical thresholds, change the pre-specified QSQ seed set or amplitude, or turn a non-evaluable material-threshold pair into a codec pass.

Supplementary Table S8 | Electron-count preservation does not certify Bader-charge fidelity

S8a. Pooled negative-control matrix across all codecs

Electron-count threshold	Bader threshold	Rows preserving electron count	Rows failing Bader threshold despite preservation	Fraction
10^{-6} e	10^{-4} e	1,108	652	58.84%
10^{-6} e	10^{-3} e	1,108	108	9.75%
10^{-6} e	10^{-2} e	1,108	10	0.90%
10^{-5} e	10^{-4} e	1,993	1,486	74.56%
10^{-5} e	10^{-3} e	1,993	482	24.18%
10^{-5} e	10^{-2} e	1,993	73	3.66%
10^{-4} e	10^{-4} e	3,205	2,691	83.96%
10^{-4} e	10^{-3} e	3,205	1,383	43.15%
10^{-4} e	10^{-2} e	3,205	306	9.55%
10^{-3} e	10^{-4} e	4,380	3,866	88.26%
10^{-3} e	10^{-3} e	4,380	2,477	56.55%
10^{-3} e	10^{-2} e	4,380	801	18.29%

S8b. Codec-resolved headline control at $|\Delta N_e| < 10^{-4}$ e and Bader error $\geq 10^{-3}$ e

Codec	Rows preserving electron count	Rows still failing Bader threshold	Fraction
ZFP	1,509	607	40.23%
SZ3	466	79	16.95%
SPERR	1,230	697	56.67%
All codecs	3,205	1,383	43.15%

Interpretation. Global electron conservation is a negative control, not a sufficient certificate for atom-resolved Bader fidelity. The main-text 3,205 / 1,383 = 43.15% result is the pooled row highlighted above.

Evidence provenance: electron-count/Bader decoupling analysis.

Supplementary Table S9 | Hartree-potential response is smoother than re-derived Bader response on the same reconstructions

S9a. Pooled codec-level scaling on the 6,270 reproduction-gate-passing rows

Codec	Gate-passing rows	Hartree slope	Hartree R^2	Bader slope	Bader R^2	Median Hartree relative RMSE	Median Bader error (e)
ZFP	2,450	0.968	0.863	0.646	0.684	9.405e-07	2.453e-03
SZ3	1,864	1.069	0.910	0.584	0.757	2.591e-05	5.890e-03
SPERR	1,956	0.988	0.786	0.587	0.750	2.643e-06	5.631e-03

The all-codec pooled Hartree log–log exponent in the pooled analysis is **1.02** on all 6,270 gate-passing rows; the per-codec values above show the same near-first-order pattern without collapsing codec-specific prefactors.

S9b. Material-level smoothness for 678 material–codec pairs with at least five gate-passing rows

Codec	Pairs	Hartree monotone	Bader monotone	Median Hartree R^2	Median Bader R^2	Median Hartree slope	Median Bader slope
ZFP	241	85.1%	19.5%	0.997	0.892	1.05	0.58
SZ3	219	85.8%	42.0%	0.994	0.947	1.17	0.60
SPERR	218	95.4%	37.2%	0.996	0.936	1.01	0.59
All	678	88.6%	32.4%	0.996	0.928	—	—

S9c. Rung-level irregularity and matched-Hartree dispersion

Diagnostic	Hartree	Bader
Local log–log elasticity range	-2.0 to +4.4	-9.3 to +14.1
Largest consecutive Bader-error jump	—	22,296×
Gate-passing rows in matched-Hartree bins with Bader P90/P10 ≥ 10	—	3,452/6,229 = 55.4%

Interpretation. The Hartree control is close to first-order at pooled and material levels, whereas the re-derived Bader response shows substantially more rung-level non-monotonicity and dispersion. The slab monotonicity caveat remains: strict Hartree monotonicity is weaker on slabs even though material-level Hartree R² remains high. The result should therefore be described as a smoother response, not as universal monotonicity.

Evidence provenance: Hartree pooled-scaling, material-smoothness and matched-error-dispersion analyses.

Interpretation. Tables S8–S9 are operator-control analyses. They show why downstream observables must be evaluated explicitly; the prospective QSQ validation remains a separate endpoint.

Supplementary Table S10 | Representative Bader decomposition confirms a dominant domain-migration channel

Relative codec tolerance	Codec	Cases	Median fixed-basin underestimation	P10–P90	Median bounded domain share at max-total atom	Domain term > integrand	Median reassigned voxel fraction
10 ⁻⁴	ZFP	12	1011.31×	87.56–11538.89×	0.999	100.0%	1.610e-03
10 ⁻⁴	SZ3	12	20.11×	3.83–30.07×	0.985	100.0%	1.216e-02
10 ⁻⁴	SPERR	12	585.30×	118.26–1147.15×	0.999	100.0%	1.945e-02
10 ⁻³	ZFP	12	171.84×	27.16–1122.03×	0.997	100.0%	5.695e-03
10 ⁻³	SZ3	12	4.00×	2.38–28.80×	0.942	91.7%	5.119e-02
10 ⁻³	SPERR	12	158.98×	46.34–466.47×	0.998	100.0%	1.182e-01
10 ⁻²	ZFP	10	43.67×	13.15–327.91×	0.996	100.0%	3.758e-02
10 ⁻²	SZ3	12	3.07×	1.26–6.83×	0.847	91.7%	1.376e-01
10 ⁻²	SPERR	12	40.29×	10.15–229.09×	0.993	100.0%	2.365e-01

Across the full representative matrix (n=106 material–codec–tolerance cases), the median maximum-resolved / maximum-fixed-basin error ratio is **52.59×**. At the atom of maximum total deviation, the median bounded absolute domain share |domain|/(|domain|+|integrand|) is **0.995**, and the domain term exceeds the integrand term in **98.1%** of cases. These two metrics are reported separately because signed cancellation can make |domain|/|total| exceed 1.

Evidence provenance: representative Bader mechanism decomposition and per-atom decomposition tables.

Supplementary Table S11 | Independent Bader implementations preserve the mechanism and strict-response ordering

S11a. Study completeness and solver failures

Quantity	Result
Expected outcome rows	1,560
Completed outcome rows	1,560
BaderKit on-grid solver failures	2
Henkelman on-grid solver failures	0
Henkelman near-grid solver failures	0
Representative systems in aggregate	12 development systems
Separate probe-failure sentinel	1 system; excluded from aggregates

S11b. Cross-implementation stability and codec response

Diagnostic	Comparison solver	n	Median / fraction	IQR
Codec response comparison/BaderKit	Henkelman on-grid (above print resolution)	70	1.000	0.920–1.005
Codec response comparison/BaderKit	Henkelman on-grid (above print and baseline compatible)	47	1.000	1.000–1.000
Codec response comparison/BaderKit	Henkelman near-grid (above print resolution)	70	0.768	0.310–1.218

Diagnostic	Comparison solver	n	Median / fraction	IQR
QSQ floor agrees with BaderKit within 10^{-6} e	Henkelman on-grid	12 materials	75.0%	—
Eligibility concordance at 0.0001 e	Henkelman on-grid	12 materials	100.0%	—
Eligibility concordance at 0.001 e	Henkelman on-grid	12 materials	100.0%	—
Eligibility concordance at 0.01 e	Henkelman on-grid	12 materials	91.7%	—
QSQ floor agrees with BaderKit within 10^{-6} e	Henkelman near-grid	12 materials	0.0%	—
Eligibility concordance at 0.0001 e	Henkelman near-grid	12 materials	83.3%	—
Eligibility concordance at 0.001 e	Henkelman near-grid	12 materials	91.7%	—
Eligibility concordance at 0.01 e	Henkelman near-grid	12 materials	91.7%	—

Codec-response ratios above use successful paired outputs above the $2e-6$ e Henkelman print-resolution region. The additional on-grid baseline-compatible row requires the unperturbed Henkelman/BaderKit atomic charges to agree within 10^{-3} e. All paired points, including flagged baseline differences, are retained in the machine-readable implementation-transfer audit.

S11c. Relative 10^{-4} codec-response ordering by implementation

Solver	Error ratio	n materials	Median ratio	IQR
BaderKit on-grid	SZ3/ZFP	12	6.18×	3.35–12.81×
BaderKit on-grid	SPERR/ZFP	12	6.69×	2.65–13.36×
Henkelman on-grid	SZ3/ZFP	12	4.83×	2.54–6.97×
Henkelman on-grid	SPERR/ZFP	12	5.45×	1.32–6.95×
Henkelman near-grid	SZ3/ZFP	12	3.17×	1.23–10.62×
Henkelman near-grid	SPERR/ZFP	12	2.87×	0.92–8.52×

S11d. Exact BaderKit decomposition by perturbation kind

Perturbation kind	n	Median bounded domain share	IQR	Domain > integrand	Max closure residual (e)
codec	70	0.9898	0.9573–0.9991	100.0%	0.0e+00
order-preserving control	12	0.0208	0.0096–0.9999	41.7%	0.0e+00
QSQ perturbation	60	0.9999	0.9990–1.0000	83.3%	0.0e+00
spatial control	324	0.9961	0.9843–0.9995	100.0%	0.0e+00

S11e. Spatial reorganization controls

Solver	Control	n pairs	Median log2(control/codec)	IQR	Fraction increased
BaderKit on-grid	global	36	0.26	-0.17–0.81	52.8%
BaderKit on-grid	shift	36	0.02	-0.29–0.63	50.0%
BaderKit on-grid	stratified	36	0.26	-0.07–0.72	63.9%
Henkelman on-grid	global	36	0.24	-0.14–0.79	61.1%
Henkelman on-grid	shift	36	0.04	-0.17–0.66	55.6%
Henkelman on-grid	stratified	36	0.15	-0.15–0.69	61.1%
Henkelman near-grid	global	36	0.07	-0.36–0.74	50.0%
Henkelman near-grid	shift	36	0.00	-0.27–0.46	50.0%
Henkelman near-grid	stratified	36	0.07	-0.24–0.47	55.6%

S11f. Resolved 24-system QSQ classification transfer

This second, deterministic panel is distinct from the 12-system mechanism matrix above. It contains 24 development systems stratified by bulk/slab and four QSQ stability-floor bands. The same five pre-specified perturbation fields were analyzed with independent Bader implementations.

Bader threshold	Independent solver	Comparable systems	Ambiguous	Agreement with primary QSQ classification	Cohen kappa	Eligible -> rejected	Rejected -> eligible
10 ⁻⁴ e	Henkelman on-grid	24	0	100.0%	1.000	0	0
10 ⁻⁴ e	Henkelman near-grid	23	1	82.6%	0.593	1	3
10 ⁻³ e	Henkelman on-grid	24	0	100.0%	1.000	0	0
10 ⁻³ e	Henkelman near-grid	24	0	83.3%	0.667	0	4
10 ⁻² e	Henkelman on-grid	24	0	100.0%	1.000	0	0
10 ⁻² e	Henkelman near-grid	24	0	95.8%	0.882	0	1

Floor-rank Spearman correlation with primary BaderKit is **0.995** for Henkelman on-grid and **0.754** for Henkelman near-grid. Recreated BaderKit floors agree with their reference values to a maximum absolute difference of **8.71338e-11 e**.

Evidence provenance: resolved 24-system implementation-transfer classification and floor-rank summaries.

Interpretation. This deliberately stratified 12-system panel is a mechanism/implementation robustness study, not a prevalence estimate. Henkelman on-grid reproduces the BaderKit codec response essentially one-for-one above print resolution, and the strict SZ3/ZFP and SPERR/ZFP ordering remains qualitatively similar under on-grid and near-grid Henkelman analyses. The exact decomposition supports a Bader-specific domain-migration channel. Spatial permutations provide secondary evidence that error organization matters, while the near-null periodic-shift control argues against a simple alignment-only explanation. These results do not redefine QSQ or alter the primary benchmark.

Evidence provenance: independent-Bader outcome, stability-comparison, mechanism-decomposition and spatial-control summaries.

Supplementary Table S12 | Realized-L[∞] matching sensitivity across pre-specified calipers

The primary analysis uses a 0.10-dex caliper; 0.05, 0.20 and 0.30 dex are pre-specified sensitivity analyses.

S12a. Matched support and realized-L[∞] balance

Caliper (dex)	Codec pair	Matched row pairs	Materials represented	Median	Δlog10 L [∞]	(dex)	P95
0.05	ZFP/SZ3	198	147	0.0234	0.0466	1.055	1.113
0.05	ZFP/SPERR	208	143	0.0256	0.0467	1.061	1.113
0.05	SZ3/SPERR	1,848	254	0.0000	0.0005	1.000	1.001
0.10	ZFP/SZ3	457	214	0.0560	0.0964	1.138	1.249
0.10	ZFP/SPERR	465	206	0.0553	0.0967	1.136	1.249
0.10	SZ3/SPERR	1,848	254	0.0000	0.0005	1.000	1.001
0.20	ZFP/SZ3	1,089	244	0.1130	0.1904	1.297	1.550
0.20	ZFP/SPERR	1,091	245	0.1124	0.1903	1.296	1.550
0.20	SZ3/SPERR	1,848	254	0.0000	0.0005	1.000	1.001
0.30	ZFP/SZ3	1,621	248	0.1464	0.2707	1.401	1.865
0.30	ZFP/SPERR	1,622	248	0.1452	0.2709	1.397	1.866
0.30	SZ3/SPERR	1,848	254	0.0000	0.0005	1.000	1.001

S12b. Re-derived Bader-error effect after matching

Caliper (dex)	Codec-error ratio	Usable matched pairs	Usable materials	Effect	95% material-bootstrap CI
0.05	ZFP/SZ3	198	147	0.583	0.549–0.650
0.05	ZFP/SPERR	208	143	0.602	0.508–0.707
0.05	SZ3/SPERR	1,848	254	1.033	0.976–1.072
0.10	ZFP/SZ3	457	214	0.557	0.525–0.598
0.10	ZFP/SPERR	465	206	0.601	0.534–0.662
0.10	SZ3/SPERR	1,848	254	1.033	0.976–1.072
0.20	ZFP/SZ3	1,089	244	0.591	0.565–0.620
0.20	ZFP/SPERR	1,091	245	0.591	0.558–0.624

Caliper (dex)	Codec-error ratio	Usable matched pairs	Usable materials	Effect	95% material-bootstrap CI
0.20	SZ3/SPERR	1,848	254	1.033	0.976–1.072
0.30	ZFP/SZ3	1,621	248	0.599	0.568–0.624
0.30	ZFP/SPERR	1,622	248	0.600	0.574–0.651
0.30	SZ3/SPERR	1,848	254	1.033	0.976–1.072

S12c. Fixed-basin diagnostic after the same matching

Caliper (dex)	Codec-error ratio	Effect	95% material-bootstrap CI
0.05	ZFP/SZ3	0.055	0.049–0.063
0.05	ZFP/SPERR	0.699	0.628–0.817
0.05	SZ3/SPERR	10.775	10.115–11.418
0.10	ZFP/SZ3	0.057	0.052–0.061
0.10	ZFP/SPERR	0.709	0.665–0.801
0.10	SZ3/SPERR	10.775	10.115–11.418
0.20	ZFP/SZ3	0.064	0.059–0.070
0.20	ZFP/SPERR	0.727	0.667–0.819
0.20	SZ3/SPERR	10.775	10.115–11.418
0.30	ZFP/SZ3	0.068	0.061–0.075
0.30	ZFP/SPERR	0.726	0.660–0.794
0.30	SZ3/SPERR	10.775	10.115–11.418

Primary 0.10-dex result. Re-derived Bader-error ratios are ZFP/SZ3 = **0.557** (95% CI 0.525–0.598), ZFP/SPERR = **0.601** (0.534–0.662), and SZ3/SPERR = **1.033** (0.976–1.072). The direction of the ZFP comparisons remains below one across all pre-specified calipers, whereas SZ3/SPERR remains close to one.

Interpretation. Matching controls realized maximum perturbation, not the full geometry of the codec error field. A residual codec effect after matching is consistent with an error-structure contribution but does not identify a unique spatial invariant. Fixed-basin effects are diagnostic because fixed domains change the downstream operator.

Evidence provenance: realized-distortion matching diagnostics and caliper-sensitivity summary.

Supplementary Table S13 | Stability-qualified rate–fidelity in development and untouched external confirmation

Development and external summaries use the same pre-specified QSQ eligibility and codec-scoring rules.

S13a. Development cohort — overall eligible material set

Bader contract	Codec	Admitted	Non-evaluable	Certified	Certified fraction	Median best-certified compression ratio [95% bootstrap CI]	P10–P90
10 ⁻⁴ e	ZFP	46	208	46	100.0%	7.8× [7.2, 8.1]	5.9–10.6×
10 ⁻⁴ e	SZ3	46	208	38	82.6%	6.4× [6.0, 7.3]	4.8–8.4×
10 ⁻⁴ e	SPERR	46	208	39	84.8%	4.2× [4.1, 4.6]	3.8–13.4×
10 ⁻³ e	ZFP	143	111	142	99.3%	13.8× [13.1, 14.8]	9.8–21.6×
10 ⁻³ e	SZ3	143	111	136	95.1%	13.2× [12.2, 14.3]	8.2–20.2×
10 ⁻³ e	SPERR	143	111	137	95.8%	5.8× [5.6, 6.0]	4.5–27.6×
10 ⁻² e	ZFP	229	25	225	98.3%	31.9× [29.9, 34.0]	19.8–52.6×
10 ⁻² e	SZ3	229	25	207	90.4%	57.7× [49.7, 66.5]	26.2–109.8×
10 ⁻² e	SPERR	229	25	208	90.8%	10.3× [9.4, 10.7]	7.1–103.1×

S13b. Development cohort — bulk/slab strata

Bader contract	Stratum	Codec	Admitted	Non-evaluable	Certified fraction	Median ratio [95% bootstrap CI]
10 ⁻⁴ e	bulk	ZFP	42	144	100.0%	7.6× [6.9, 8.0]
10 ⁻⁴ e	bulk	SZ3	42	144	85.7%	6.3× [5.8, 7.2]
10 ⁻⁴ e	bulk	SPERR	42	144	90.5%	4.2× [4.1, 4.6]

Bader contract	Stratum	Codec	Admitted	Non-evaluable	Certified fraction	Median ratio [95% bootstrap CI]
	10 ⁻⁴ e slab	ZFP	4	64	100.0%	10.5× [9.1, 11.6]
	10 ⁻⁴ e slab	SZ3	4	64	50.0%	8.4× [8.3, 8.5]
	10 ⁻⁴ e slab	SPERR	4	64	25.0%	3.9× [3.9, 3.9]
	10 ⁻³ e bulk	ZFP	103	83	99.0%	13.5× [12.8, 14.5]
	10 ⁻³ e bulk	SZ3	103	83	99.0%	12.9× [12.0, 14.3]
	10 ⁻³ e bulk	SPERR	103	83	99.0%	6.2× [6.0, 6.5]
	10 ⁻³ e slab	ZFP	40	28	100.0%	15.3× [13.0, 18.4]
	10 ⁻³ e slab	SZ3	40	28	85.0%	14.1× [10.8, 15.4]
	10 ⁻³ e slab	SPERR	40	28	87.5%	5.1× [4.9, 5.4]
	10 ⁻² e bulk	ZFP	168	18	98.8%	30.0× [27.1, 31.9]
	10 ⁻² e bulk	SZ3	168	18	95.2%	51.8× [47.7, 60.3]
	10 ⁻² e bulk	SPERR	168	18	95.8%	11.5× [10.1, 13.2]
	10 ⁻² e slab	ZFP	61	7	96.7%	40.5× [35.8, 44.9]
	10 ⁻² e slab	SZ3	61	7	77.0%	67.8× [59.6, 70.6]
	10 ⁻² e slab	SPERR	61	7	77.0%	8.1× [7.7, 8.6]

S13c. Untouched external confirmatory cohort — overall

Bader contract	Codec	Admitted (of 63)	Non-evaluable	Certified	Certified fraction	Median best-certified compression ratio [95% bootstrap CI]	Row-failure-affected materials
10 ⁻⁴ e	ZFP	16	47	16	100.0%	13.0× [7.1, 20.4]	0
10 ⁻⁴ e	SZ3	16	47	14	87.5%	12.1× [7.3, 17.7]	0
10 ⁻⁴ e	SPERR	16	47	14	87.5%	5.2× [4.2, 6.2]	0
10 ⁻³ e	ZFP	42	21	42	100.0%	18.8× [14.3, 25.0]	0
10 ⁻³ e	SZ3	42	21	42	100.0%	18.8× [12.1, 24.6]	0
10 ⁻³ e	SPERR	42	21	41	97.6%	6.4× [5.9, 6.9]	0
10 ⁻² e	ZFP	57	6	57	100.0%	40.6× [35.4, 46.0]	2
10 ⁻² e	SZ3	57	6	57	100.0%	65.9× [40.7, 101.8]	0
10 ⁻² e	SPERR	57	6	57	100.0%	10.8× [10.2, 12.4]	0

S13d. Untouched external cohort — bulk/vacuum-containing 2D strata

Bader contract	Stratum	Codec	Admitted	Non-evaluable	Certified fraction	Median ratio [95% bootstrap CI]
	10 ⁻⁴ e bulk	ZFP	9	27	100.0%	7.1× [6.3, 20.2]
	10 ⁻⁴ e bulk	SZ3	9	27	77.8%	7.3× [4.5, 17.4]
	10 ⁻⁴ e bulk	SPERR	9	27	88.9%	4.3× [3.9, 11.1]
	10 ⁻⁴ e vacuum-containing 2D	ZFP	7	20	100.0%	19.7× [14.9, 27.8]
	10 ⁻⁴ e vacuum-containing 2D	SZ3	7	20	100.0%	17.1× [9.9, 21.5]
	10 ⁻⁴ e vacuum-containing 2D	SPERR	7	20	85.7%	5.5× [4.7, 6.0]
	10 ⁻³ e bulk	ZFP	21	15	100.0%	11.2× [9.7, 13.9]
	10 ⁻³ e bulk	SZ3	21	15	100.0%	9.8× [8.2, 16.8]
	10 ⁻³ e bulk	SPERR	21	15	100.0%	6.2× [5.0, 7.0]
	10 ⁻³ e vacuum-containing 2D	ZFP	21	6	100.0%	25.6× [24.8, 30.2]
	10 ⁻³ e vacuum-containing 2D	SZ3	21	6	100.0%	24.6× [22.9, 39.1]
	10 ⁻³ e vacuum-containing 2D	SPERR	21	6	95.2%	6.6× [6.0, 7.0]
	10 ⁻² e bulk	ZFP	30	6	100.0%	28.5× [23.1, 33.5]
	10 ⁻² e bulk	SZ3	30	6	100.0%	33.0× [26.5, 39.6]
	10 ⁻² e bulk	SPERR	30	6	100.0%	13.1× [9.8, 22.8]
	10 ⁻² e vacuum-containing 2D	ZFP	27	0	100.0%	49.1× [42.4, 61.1]
	10 ⁻² e vacuum-containing 2D	SZ3	27	0	100.0%	115.2× [101.8, 158.4]

Bader contract	Stratum	Codec	Admitted	Non-evaluable	Certified fraction	Median ratio [95% bootstrap CI]
10 ⁻² e	vacuum-containing 2D	SPERR	27	0	100.0%	10.2× [9.6, 11.6]

S13e. Pairwise SZ3 versus ZFP transition

Cohort	Bader contract	Stratum	n eligible comparisons	SZ3 wins	Ties	ZFP wins	Median log2(SZ3/ZFP)
development	10 ⁻⁴ e	overall	46	10.9%	0.0%	89.1%	-0.199
development	10 ⁻⁴ e	bulk	42	11.9%	0.0%	88.1%	-0.187
development	10 ⁻⁴ e	slab	4	0.0%	0.0%	100.0%	-0.392
development	10 ⁻³ e	overall	143	32.9%	0.0%	67.1%	-0.106
development	10 ⁻³ e	bulk	103	40.8%	0.0%	59.2%	-0.095
development	10 ⁻³ e	slab	40	12.5%	0.0%	87.5%	-0.290
development	10 ⁻² e	overall	229	80.3%	1.3%	18.3%	0.746
development	10 ⁻² e	bulk	168	86.9%	0.6%	12.5%	0.869
development	10 ⁻² e	slab	61	62.3%	3.3%	34.4%	0.571
external confirmatory	10 ⁻⁴ e	overall	16	31.2%	0.0%	68.8%	-0.148
external confirmatory	10 ⁻⁴ e	bulk	9	22.2%	0.0%	77.8%	-0.122
external confirmatory	10 ⁻⁴ e	vacuum-containing 2D	7	42.9%	0.0%	57.1%	-0.173
external confirmatory	10 ⁻³ e	overall	42	35.7%	0.0%	64.3%	-0.116
external confirmatory	10 ⁻³ e	bulk	21	33.3%	0.0%	66.7%	-0.173
external confirmatory	10 ⁻³ e	vacuum-containing 2D	21	38.1%	0.0%	61.9%	-0.081
external confirmatory	10 ⁻² e	overall	57	89.5%	0.0%	10.5%	0.646
external confirmatory	10 ⁻² e	bulk	30	83.3%	0.0%	16.7%	0.299
external confirmatory	10 ⁻² e	vacuum-containing 2D	27	96.3%	0.0%	3.7%	1.316

External confirmation summary. The untouched cohort admits 16/63, 42/63 and 57/63 systems at 10⁻⁴, 10⁻³ and 10⁻² e. Median external best-certified compression ratios are approximately ZFP/SZ3/SPERR = 13.0/12.1/5.2× at 10⁻⁴ e, 18.8/18.8/6.4× at 10⁻³ e, and 40.6/65.9/10.8× at 10⁻² e. The SZ3-versus-ZFP win fraction correspondingly shifts from 31.3% to 35.7% to 89.5%.

Interpretation. The preferred codec depends on the scientific contract and the eligible cohort. These tables do not support a universal codec winner. The 63-system confirmatory cohort must not be replaced by the 65-system descriptive/stability aggregate when reporting external rate-fidelity.

Evidence provenance: development/external rate-fidelity summary and SZ3/ZFP pairwise comparison table.

Supplementary Table S14 | Failure taxonomy, exclusions and negative-result audit

S14a. Failure and exclusion semantics used in the submission

State / category	Meaning	Scored as codec pass/fail?	Evidence class
Reference-QoI non-evaluability	Reference Bader QoI fails QSQ eligibility at the requested tolerance	No — neither pass nor fail	QSQ eligibility summary
Bader solver failure	Downstream Bader analysis did not return a valid row-level result	No automatic codec attribution; retained in audit	failure registry
Reproduction mismatch	Reproduction/platform check failed despite reconstructed scientific field remaining consistent	No; infrastructure exclusion from the affected formal analysis	Hartree reproduction records
Symmetry-equivalent basin relabelling	Apparent large atom-indexed charge change is a permutation among symmetry-equivalent basins	Non-evaluable for position-indexed charge, separately flagged	failure registry
eligible + not certified	Reference QoI is numerically eligible but compressed reconstruction exceeds the requested Bader contract	Yes: genuine failure	development benchmark
eligible + certified	Reference QoI is eligible and the reconstruction satisfies the contract	Yes: certified success	development benchmark

S14b. Explicit failure registry counts

Registry category	Row records
Bader solver failure	78
Symmetry-equivalent basin relabelling	1

Bader-solver failures by corpus and codec

Corpus	Codec	Row failures
development bulk	ZFP	37
development slab	SPERR	4
development slab	ZFP	37

These counts describe explicit row-level failure records, not material-level non-evaluatable counts. QSQ non-evaluability is reported separately through the eligibility analysis and should not be reconstructed from the failure registry.

S14c. Exploratory algorithm checks not retained as benchmark claims

Check	Result / reason not promoted
External baselines (BQB, den2bin)	Retained as related-work context rather than headline fair benchmarks because task definitions and error-control contracts are not directly comparable to the density-field benchmark.
Candidate codec modifications	No proposed modification showed sufficiently robust improvement under the re-derived-Bader metric to support a new-compressor claim; the contribution therefore remains the measurement/certification framework.

Interpretation. Scientific non-evaluability, downstream solver failure, infrastructure mismatch and codec fidelity are reported as distinct states. Non-evaluatable targets are not codec failures, and solver failures are not silently imputed as codec failures.

Evidence provenance: failure-registry summary and claim–evidence audit.

Supplementary Table S15 | Outcome-blind chemistry pairs and two-implementation source references

Pair	Source chemistry	State A → State B	Added atoms	Target	Δq BaderKit (e)	Δq Henkelman (e)	Pair QSQ eligible at 10^{-3} e
Pair 1	GaN electrochemical surfaces, with AECCARs	Ga22N22 → Ga22N22H2H2	H:4	N	0.528936	0.528936	Yes
Pair 2	GaN electrochemical surfaces, with AECCARs	Ga15N15H2H1 → Ga15N15H3H3	H:3	N	0.522669	0.522669	Yes
Pair 3	GaN electrochemical surfaces, with AECCARs	Ga15N15 → Ga15N15H2H1	H:3	N	0.610388	0.610388	Yes
Pair 4	RuO ₂ CO ₂ RR, adsorbate and spectator variations	O31Ru16 → H3C1O31Ru16	C:1;H:3	O	0.421202	0.421201	No
Pair 5	RuO ₂ CO ₂ RR, adsorbate and spectator variations	Ru16C3O33 → Ru16C3O34H1	H:1;O:1	Ru	-0.131815	-0.131814	No

Reference acceptance was pre-specified before compression outcomes: identical non-zero sign in BaderKit and Henkelman on-grid, minimum $|\Delta q| \geq 0.02$ e, and inter-solver $|\Delta q|$ disagreement ≤ 0.01 e. All five pairs passed.

Supplementary Table S16 | Direct-selection and optional independent-solver escalation policies

Direct decision policies, pooled across codecs

Policy	Retained / valid	Coverage	Adverse or zero-direction decisions	Error rate	Unique retained pairs
Unqualified baseline	60 / 60	100.0%	0	0.0%	5
Realized- L^∞ coverage-matched	36 / 60	60.0%	0	0.0%	5
Order-preserving control	60 / 60	100.0%	0	0.0%	5
QSQ	36 / 60	60.0%	0	0.0%	3

Optional independent-solver escalation

Policy	Resolved / valid	Needs review	Errors among resolved	Henkelman state calls	Henkelman wall time (s)
QSQ-targeted escalation	60 / 60	0	0	48	397.2

Policy	Resolved / valid	Needs review	Errors among resolved	Henkelman state calls	Henkelman wall time (s)
Escalate all	60 / 60	0	0	108	733.3

Interpretation: no policy improves sign correctness because the unqualified common-tight baseline already has zero adverse decisions. QSQ is more conservative than this coarse sign contract. The lower independent-solver call count for targeted escalation is an audit-cost comparison only.

Supplementary Table S17 | Operator-resolved Fourier diagnostics for realized- L^∞ -matched ZFP/SZ3 reconstructions.

The mechanism population contains 457 matched pairs from 214 materials (914 reconstructions). Material-level centers are computed after within-material aggregation on the log scale unless otherwise stated. The exact multiplicative Hartree decomposition is evaluated pairwise; separately aggregated centers are descriptive and are not expected to multiply exactly.

Diagnostic	Value	Interpretation
Reference-implementation Hartree error ratio, ZFP/SZ3	0.0776221	Reproduces the matched-pair Hartree effect used as the mechanism target
Nyquist-safe Hartree error ratio, ZFP/SZ3	0.0776219	Shows that the codec effect is unchanged after restoring exact Hermitian parity
Total spectral-error-energy factor, $\sqrt{(E\text{ ZFP}/E\text{ SZ3})}$	0.376	Contribution from overall spectral error magnitude
Spectral Hartree-susceptibility factor, $\sqrt{(S\text{ H,ZFP}/S\text{ H,SZ3})}$	0.203	Contribution from frequency allocation under Hartree weighting
Material-median spectral-structure share of absolute log effect	62.0%	Frequency allocation is the larger component of the matched-distortion Hartree effect
Low-G error-energy fraction ratio, ZFP/SZ3	0.416	ZFP places less reconstruction-error energy in long-wavelength modes
High-G error-energy fraction ratio, ZFP/SZ3	0.626	ZFP also differs at high G, but these modes are weakly weighted by the Hartree operator
Materials with lower ZFP spectral Hartree susceptibility	99.5%	Direction of the susceptibility effect is nearly universal in the mechanism cohort
Materials with higher ZFP spectral centroid	98.1%	ZFP error is shifted toward higher reciprocal-space frequencies
Materials with lower ZFP low-G error-energy fraction	99.1%	Long-wavelength suppression is consistent across materials
Maximum real-space/Fourier-space Parseval relative discrepancy	1.30×10^{-15}	Confirms numerical identity of the Nyquist-safe operator and spectral expression

For the Nyquist-safe reciprocal-space set G_s ,

$$E=\sum_{G\in G_s}|\Delta\rho(G)|^2, WH=\sum_{G\in G_s}(|\Delta\rho(G)|^2)/(|G|^4), SH=(WH)/(E).$$

For every matched pair,

$$RH=\sqrt{((E\text{ ZFP})/(E\text{ SZ3}))}\sqrt{((S\text{ H,ZFP})/(S\text{ H,SZ3}))}.$$

Machine-readable evidence: Fourier-spectrum mechanism audit in the repository reader-facing provenance index, including the matched-pair mechanism table, reconstruction spectral metrics, radial-spectrum summary, mechanism-ratio summary and audit summary.